

Algebra
Home Work 4

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A) Let $\ell \geq 1$ be an integer and consider the cyclic group $(\mathbb{Z}/\ell\mathbb{Z}, +)$. Given $n \in \mathbb{Z}$, we write $\bar{n} = n + \ell\mathbb{Z}$ for the class of n in $\mathbb{Z}/\ell\mathbb{Z}$.

- (1) Show that there is a well-defined composition law $\times$ on $\mathbb{Z}/\ell\mathbb{Z}$ such that $\overline{m \times n} = \overline{m} \times \overline{n}$, for all $m, n \in \mathbb{Z}$.
- (2) Show that $(\mathbb{Z}/\ell\mathbb{Z}, \times)$ is a commutative monoid.
- (3) For $n \in \mathbb{Z}$, show that $\bar{n}$ is invertible in $(\mathbb{Z}/\ell\mathbb{Z}, \times)$ iff $\gcd(n, \ell) = 1$.
- (4) Let $(\mathbb{Z}/\ell\mathbb{Z})^\times \subset \mathbb{Z}/\ell\mathbb{Z}$ be the subset of invertible elements in $(\mathbb{Z}/\ell\mathbb{Z}, \times)$. Show that $((\mathbb{Z}/\ell\mathbb{Z})^\times, \times)$ is an abelian group.
- (5) Let p be a prime number. Show that $\bar{n}^{p-1} = \bar{1}$ in $\mathbb{Z}/p\mathbb{Z}$ for all $1 \leq n \leq p-1$. (*Hint:* Note that $(\mathbb{Z}/p\mathbb{Z})^\times = (\mathbb{Z}/p\mathbb{Z}) \setminus \{\bar{0}\}$ and use the previous question.)
- (6) Deduce that for every $x \in \mathbb{Z}$, $x^p - x$ is divisible by p . (This is Fermat's little theorem.)

Answer:

- (1) For $(\mathbb{Z}/l\mathbb{Z}, \times)$, $l \geq 1$, to be well defined we need to show that:

$$\overline{m \times n} = \overline{m} \times \overline{n} \quad \forall m, n \in \mathbb{Z}$$

is independent of any of the representatives taken from $\overline{m}$ and $\overline{n}$.

Let: $n_1, n_2 \in \bar{n}$ and $m_1, m_2 \in \overline{m}$, then we must have there are $q_1, q_2, p_1, p_2 \in \mathbb{Z}$ such that: $n_1 = n + q_1l$, $n_2 = n + q_2l$, $m_1 = m + p_1l$ and $m_2 = m + p_2l$. Then:

$$n_1 \times m_1 = (n + q_1l)(m + p_1l) = nm + r_1l \quad (\text{where } r_1 \in \mathbb{Z}) \quad (1.1)$$

$$n_2 \times m_2 = (n + q_2l)(m + p_2l) = nm + r_2l \quad (\text{where } r_2 \in \mathbb{Z}) \quad (1.2)$$

Both representations are in $\overline{m \times n}$ and so the operation is well defined.

- (2) To show that $(\mathbb{Z}/l\mathbb{Z}, \times)$ is a monoid one need only show that the composition operation $(\times)$ is associative and that there is an identity element.

Clearly $\bar{1}$ is the identity element.

Following from part (1) - To show that $\times$ is associative we take 3 elements and show that composition is associative. Let $\bar{a}, \bar{b}, \bar{c}$ be the three elements of $(\mathbb{Z}/l\mathbb{Z}, \times)$.

$$\begin{aligned} \bar{a} \times (\bar{b} \times \bar{c}) &= \bar{a} \times \overline{(b \times c)} \\ &= \overline{a \times b \times c} \end{aligned}$$

Also:

$$\begin{aligned} (\bar{a} \times \bar{b}) \times \bar{c} &= \overline{(a \times b)} \times \bar{c} \\ &= \overline{a \times b \times c} \end{aligned}$$

□

(3) Let $\bar{n} \in (\mathbb{Z}/l\mathbb{Z}, \times)$ be invertible. Then $\exists \bar{m} \in (\mathbb{Z}/l\mathbb{Z}, \times)$ such that:

$$\bar{n} \times \bar{m} = \overline{n \times m} = \bar{1}$$

Put another way, take any $n \in \bar{n}$ then this number is invertible in $(\mathbb{Z}/l\mathbb{Z}, \times) \iff \gcd(n, l) = 1$.

Proof:

Assume that n is invertible in $(\mathbb{Z}/l\mathbb{Z}, \times)$. Then there exists an $m \in (\mathbb{Z}/l\mathbb{Z}, \times)$ such that:

$$\begin{aligned} (n + k_1 l)(m + k_2 l) &= (1 + k_3 l) \\ \implies nm + k_4 l &= 1 + k_3 l \\ \implies nm + (k_4 - k_3)l &= 1 \end{aligned} \tag{1.3}$$

where $k_1, k_2, k_3, k_4 \in \mathbb{Z}$. Now let $\gcd(n, l) = d \geq 1$, then as $d|n$ and $d|l$ we can use (1.3) to determine that $d|1$. Hence $d = 1$ and n and l are coprime.

Now assume that $\gcd(n, l) = 1$. We can use Bézout's lemma to determine integers $p, q \in \mathbb{Z}$ such that:

$$pn + ql = 1$$

This means that $pn \in \bar{1}$ from which we have that n and hence $\bar{n}$ is invertible - with inverse: $\bar{p}$.

(4) We have that $(\mathbb{Z}/l\mathbb{Z})^*$ consists of all invertible elements of $(\mathbb{Z}/l\mathbb{Z}, \times)$.

Clearly $\bar{1} \in (\mathbb{Z}/l\mathbb{Z})^*$, since $\bar{1} \times \bar{1} = \bar{1}$, which is the identity element for this set.

Let $\bar{p} \in (\mathbb{Z}/l\mathbb{Z})^*$, then this element has an inverse $\bar{p}^{-1} = \overline{p^{-1}} \in (\mathbb{Z}/l\mathbb{Z})^*$ such that: $\bar{p} \times \bar{p}^{-1} = \overline{p \times p^{-1}} = \bar{1}$. But this also means that: $\overline{p^{-1}} \in (\mathbb{Z}/l\mathbb{Z})^*$.

Now let $\bar{p}, \bar{q} \in (\mathbb{Z}/l\mathbb{Z})^*$. Then the inverses, as we saw above, must also be in the set: i.e. $\bar{p}^{-1}, \bar{q}^{-1} \in (\mathbb{Z}/l\mathbb{Z})^*$.

But then: $\bar{p} \times \bar{q} = \overline{p \times q}$ will have the inverse element: $\overline{p^{-1} \times q^{-1}} = \overline{p^{-1}} \times \overline{q^{-1}} = (\overline{p \times q})^{-1}$. Hence: $\overline{p \times q} \in (\mathbb{Z}/l\mathbb{Z})^*$.

All 3 conditions for a set to be a group are established - associativity was already established earlier. Hence: $(\mathbb{Z}/l\mathbb{Z})^*$ is indeed a group. Further, as multiplication is commutative it is an abelian group.

B) Let p be a prime number. We fix a group G of cardinality p^2 . Our goal is to show that G is abelian. We argue by contradiction assuming that G is not abelian.

- (1) Show that every element of $G \setminus \{1\}$ has order p .
- (2) Let S be the set of all cyclic subgroups of G of cardinality p . Show that the set $\mathcal{P} = \{H \setminus \{1\} \mid H \in S\}$ is a partition of $G \setminus \{1\}$. Deduce that $\text{card}(S) = p + 1$.
- (3) Fix two elements x and y in G such that $xy \neq yx$. Show that $x^m y^n \neq y^n x^m$ for all $1 \leq m \leq p - 1$ and $1 \leq n \leq p - 1$.
- (4) For $0 \leq m \leq p - 1$, consider the cyclic subgroup $H_m = \langle y^{-m} x y^m \rangle$ of G . Show that $H_m = \{y^{-m} x^n y^m \mid 0 \leq n \leq p - 1\}$.
- (5) Let $0 \leq m_1 < m_2 \leq p - 1$. We want to show that $H_{m_1} \neq H_{m_2}$. We argue by contradiction and assume that $H_{m_1} = H_{m_2}$.
 - (a) Set $l = m_2 - m_1$. Show that there is an integer $0 \leq e \leq p - 1$ such that $x^e = y^{-l} x y^l$.
 - (b) Deduce by induction on $r \in \mathbb{N}$ that $y^{-lr} x y^{lr} = x^{e^r}$.
 - (c) By letting $r = p$, deduce that $x = x^{e^p}$.
 - (d) Use Exercise **A**), (6) to deduce a contradiction.
- (6) Deduce that the subset $K = (G \setminus \bigcup_{0 \leq m \leq p-1} H_m) \cup \{1\}$ is a subgroup of G of cardinality p . (Use questions (2) and (5).)
- (7) Show that $G = \langle x \rangle \langle y \rangle$. Deduce that every element of G can be written as $x^n y^m$ for some integers $0 \leq n \leq p - 1$ and $0 \leq m \leq p - 1$.
- (8) Deduce from the previous question that every conjugate of $\langle x \rangle$ is one of the H_m , for $0 \leq m \leq p - 1$. More precisely, show that for every $g \in G$ we have $\langle g^{-1} x g \rangle = H_m$ for exactly one integer $0 \leq m \leq p - 1$.
- (9) Using questions (6) and (8), show that K is a normal subgroup of G .
- (10) Let $z \in K$ be a generator. Deduce from question (9) that $x^{-1} z x = z^e$ for some $0 \leq e \leq p - 1$. Use the method of question (5) to show that $e = 1$. Obtain the final contradiction. (Note that $x \notin K$.)

Answer:

(1) If G is a group of order p^2 , where p is a prime, then the only divisors are: $1, p, p^2$. Hence the orders of the elements in G must be one of these three values. The only element of order 1 is 1 (identity element). If we have an element of order p^2 then G is cyclic and hence abelian. So, if we are to assume that G is NOT abelian, we must conclude that all elements, other than 1, will have order p .

(2) As all elements $g \in G \setminus \{1\}$ have order p the associated cyclic groups $\langle g \rangle$ also have order p .

Let us define an equivalence relation on $G \setminus \{1\}$ defined by: $g_1 \sim g_2 \iff g_2 \in \langle g_1 \rangle$.
Then:

- $g_1 \sim g_1; \forall g \in G \setminus \{1\}$
- $g_1 \sim g_2 \implies g_2 \sim g_1$,
since $g_2 = g_1^k$ for some $k \implies g_2^r = g_1$, where $rk \equiv 1 \pmod{p^2}$
- $g_1 \sim g_2$ and $g_2 \sim g_3 \implies g_1 \sim g_3$,
since $g_2 = g_1^{k_1}$ and $g_3 = g_2^{k_2} \implies g_3 = g_1^{k_1 k_2 \pmod{p^2}}$

Normally each cycle would have p elements - but actually have $(p - 1)$ as we excluded 1. Hence we have k such cycles and this must account for the $p^2 - 1$ elements in $G \setminus \{1\}$.

Counting elements we arrive at: $k(p - 1) = p^2 - 1 = (p - 1)(p + 1)$. From this we can conclude that there must be $k = p + 1$ such cycles. In the notation of the question: $\text{Card}(S) = p + 1$.

(3) As G is assumed to not be abelian there must exist distinct $x, y \in G$ such that: $xy \neq yx$.

We can proceed as follows:

$$xy \neq yx \iff y \neq xyx^{-1}$$